Attitude and Direction – General Data

1. General

The attitude and direction system is comprised of the standby magnetic compass, the Electronic Standby Instrument System (ESIS), the Air Data, Attitude and Heading Reference Systems (ADAHRS 1 and ADAHRS 2) and is displayed through the Primary Flight Display, (PFD) and the Multi-Function Displays, (MFDs).

2. Description

A. Standby Magnetic Compass

Refer to 34-25-00, Standby Magnetic Compass – General Data for system details.

The standby magnetic compass gives the aircraft magnetic heading.

B. Electronic Standby Instrument System (ESIS)

Refer to 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049 and Mod 6/2170) – General Data for system details.

The Electronic Standby Instrument System (ESIS) gives displays for attitude, heading, altitude and airspeed in case of primary display or systems failures. The ESIS is powered by its own dedicated PS–850B backup battery, and is entirely independent of the Apex system.

C. Air Data, Attitude and Heading Reference Systems (ADAHRS)

Refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data for system details.

Each channel of the ADAHRS contains a solid-state micro-electro-mechanical system (MEMS) technology sensor block, which contains three rate sensors and three accelerometers in an orthogonal triad configuration. The triad in Channel B is skewed relative to Channel A. Each channel has a connection for an Outside Air Temperature (OAT) probe, a magnetometer, and two isolated absolute pressure sensors (one for pitot and one for static pressure). Channel A receives inputs from the left pitot tube, lower static ports, and one channel of the temperature probe and magnetometer. Channel B receives inputs from the right pitot tube, upper static ports, and the second channel of the temperature probe and magnetometer. Each channel of the ADAHRS has a Central Processing Unit (CPU). The ADAHRS is installed in the nose avionics rack. A Rosemount (Goodrich) dual element OAT sensor is mounted just forward and below the ram air scoop on the left side of the fuselage to supply the ADAHRS with OAT data. Two KMG 7010 magnetometers are mounted near the outboard end of the right wing (Pre Mod 6/2171), or in the vertical stabilizer (Post Mod 6/2171), one for each channel of the ADAHRS. During normal operation, the left pilot's PFD receives ADAHRS source data from the left pitot tube and lower static ports and ADAHRS Channel A. The right pilot's PFD receives ADAHRS source data from the right pitot tube, upper static ports, and ADAHRS Channel B.

Each ADAHRS channel CPU receives air data, temperature and heading information from that channel's sensor block and passes it to the other channel. Both CPUs then compare the data to verify sensor integrity. Verified ADAHRS information is sent to each channel of the MAU for display on the Apex System.

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